

# Dual Sensor Thermal Imaging System for Forest Fire Prevention

## ANALISA JANGKAUAN PANTAUAN DUAL SENSOR THERMAL IMAGING SYSTEM UNTUK PENCEGAHAN KEBAKARAN DI PERKEBUNAN SAWIT



LOKASI PEMASANGAN  
Perkebunan Sawit

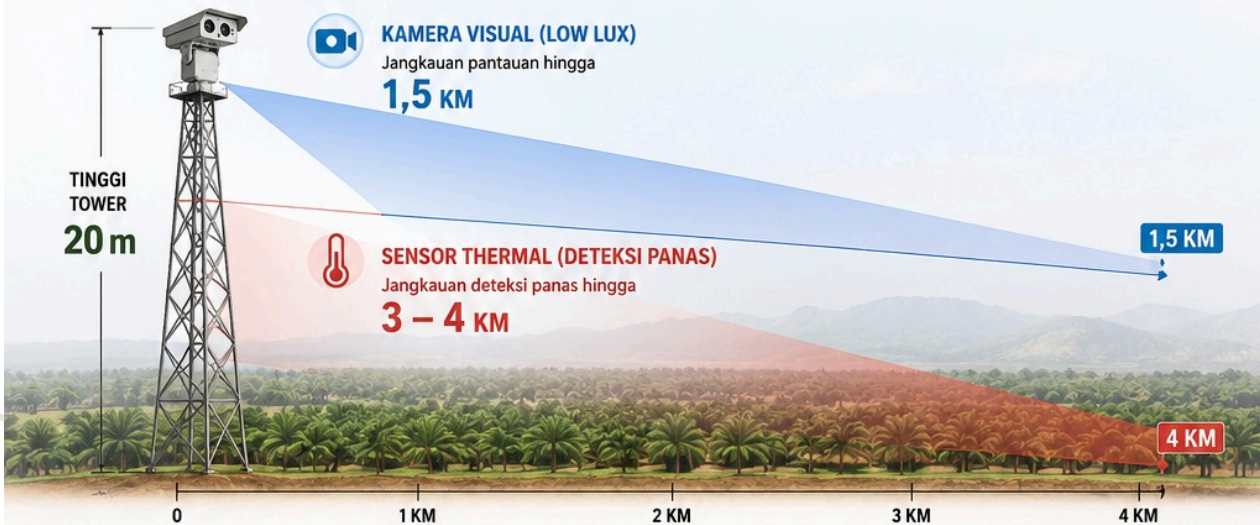


TINGGI PEMASANGAN  
Tower **20 Meter**



KETAHANAN  
**IP66**  
Tahan hujan, debu, panas  
dan korosi

### 1. ANALISA JANGKAUAN PANTAUAN



#### JANGKAUAN KAMERA VISUAL (LOW LUX)

##### DETAIL SPESIFIKASI

- Lensa 37x Optical Zoom (6.5 ~ 240mm)
- Jangkauan pengenalan objek hingga 1,5 km
- Membantu melihat detail objek/kebakaran pada siang & malam, bahkan menembus kabut

#### CONTOH CAPAIAN



500 m



1.000 m



1.500 m

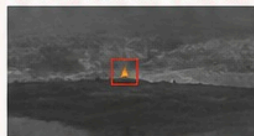


#### JANGKAUAN SENSOR THERMAL (DETEKSI PANAS)

##### DETAIL SPESIFIKASI

- Lensa 50mm (FOV 8.8° x 7.0°)
- Deteksi panas (hotspot) hingga 3 – 4 km
- Resolusi termal 640 x 480, sensitivitas  $\leq 35mK$
- Mampu mendeteksi sumber panas kecil (api awal) di area luas

#### CONTOH DETEKSI API (FIRE SIZE)



Api 1m x 1m  
Terdeteksi: 1 – 2 KM

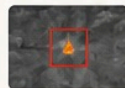


Api 2m x 2m  
Terdeteksi: 3 – 4 KM

### 2. KEMAMPUAN DETEKSI KEBAKARAN



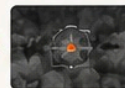
- Mendeteksi area suhu tinggi (hotspot) minimal 2x2 pixel
- Alarm otomatis ketika terdeteksi panas di atas ambang batas
- Respon alarm cepat  $\leq 0,1$  detik, tanpa delay
- Kamera otomatis men-zoom & center ke titik api
- Mendukung overlay alarm di video & koordinat titik panas



Deteksi Panas



Alarm Aktif

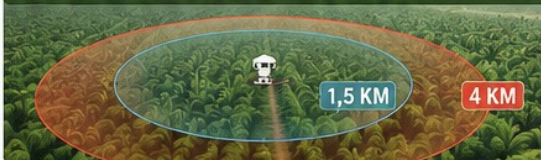


Auto Zoom



Monitoring & Verifikasi

### 3. ILUSTRASI CAKUPAN DARI TOWER 20 METER



— Jangkauan Kamera Visual  
— Jangkauan Sensor Thermal (Deteksi Panas)

### 4. CATATAN PENTING



Jangkauan dapat dipengaruhi oleh kondisi cuaca, kabut tebal, hujan lebat, dan kontur medan.



Vegetasi rapat dapat menghalangi pandangan visual, namun sensor thermal tetap dapat mendeteksi panas.



Sistem ideal dipasang dengan platform software untuk monitoring real-time dan manajemen alarm.



AMAN, CEPAT, TEPAT  
LINDUNGI PERKEBUNAN KITA

Deteksi Dini • Respon Cepat • Lindungi Aset Perkebunan

# Dual Sensor Thermal Imaging System for Forest Fire Prevention

*See through the fog, smoke, dust and other hazardous atmospheres in total darkness*



**SHR-HLVDIR50RF** HD Dual Sensor Forest Fire Prevention System integrates thermal imaging camera with temperature alarm function, visible camera, pantilt, etc. The system can achieve unattended work uninterrupted. It can analysis the video info and make the judgement automatically, timely detection the signs of fire in the surveillance area, in the fastest and best way to send the alarm, and minimize false positives; at the same time, you can check the real-time videos of the scene and directly dispatch the fire control according to the visual images, effectively assist the firefighters to deal with fire crisis.

The whole system includes camera, transmission system and display platform. The camera and control software are main designed for forest fire alarm. It can work with other device to realize alarm sound, alarm signal on the screen, etc.

## I. Features

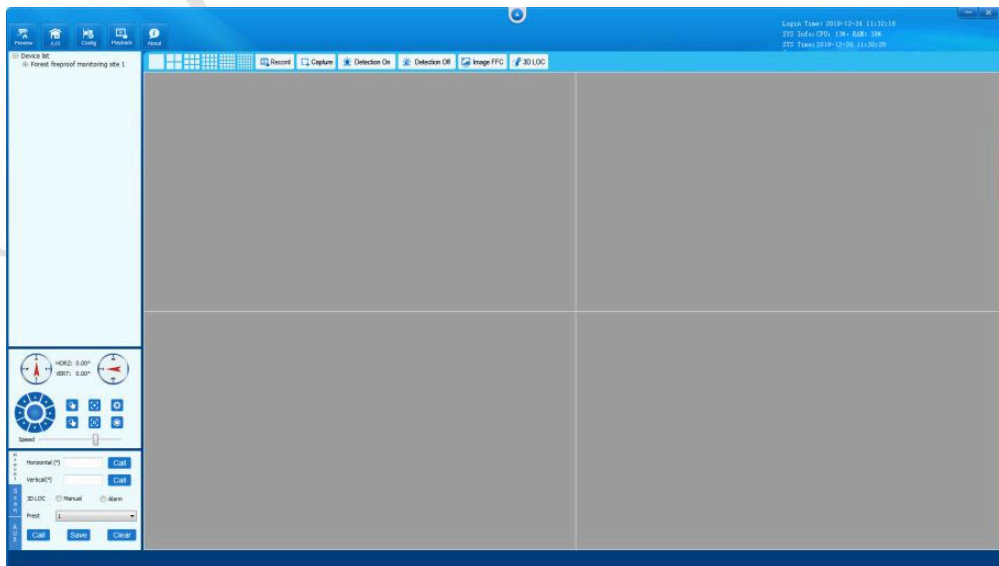
- ☐ Continuous zoom thermal imaging technology, clear imaging, detection distance, either large-scale search of the fire, but also to closely observe fire conditions
- ☐ Hot target intelligent detection alarm, high sensitivity, minimum temperature resolution of 35mK
- ☐ After the alarm, the fire source can be automatically centered zoom, easy to confirm the fire
- ☐ Low Lux day & night camera through fog which can help people observe fire details.
- ☐ Excellent Auto Focus Technology, can achieve a variety of ways to focus and trigger
- ☐ 3D Located Function, in thermal and visible camera images, can achieve marquee zoom in and click on the center function, reduce operational complexity, improve device usability
- ☐ Thermal imager presets, thermal imager and visible lens can achieve preset function, the viewing angle memorized for each preset point, when calling the preset position, the camera can automatically adjust to the memory of the viewing angle to avoid traditional thermal imager every manual adjustment, improve work efficiency
- ☐ Universal variable speed PTZ, to achieve a comprehensive no blind corner monitor, do high-precision positioning
- ☐ Professional design shell, beautiful, light, rugged, anti-high temperature, corrosion-resistant, anti-rain water, anti-acid rain, anti-salt fog

# Dual Sensor Thermal Imaging System for Forest Fire Prevention

## II. Application Show



## III. Software show



## IV. Technical Indicators

|                                |                                        |               |
|--------------------------------|----------------------------------------|---------------|
| Detection distance             |                                        |               |
| Thermal imaging alarm distance | Fire(1m x 1m)                          | Fire(2m x 2m) |
|                                | 1~2km                                  | 3~4km         |
| Visible camera                 | 1.5km                                  |               |
| Alarm detect                   |                                        |               |
| Alarm Condition                |                                        |               |
|                                | High-temperature area occupy pixel 2×2 |               |

# Dual Sensor Thermal Imaging System for Forest Fire Prevention

|                       |                                                                                                                                               |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Alarm threshold value | 1~255 grade, can adjust threshold value to set alarm sensitivity                                                                              |
| Target range          | 2×2~ 255×255 pixel                                                                                                                            |
| Alarm target          | Can set 1-10 unit alarm target, auto display over-temp target                                                                                 |
| Alarm type            | Video superposition alarm box/switching value/data returned pixel coordinates, alarm intuitive easy to read                                   |
| Information algorithm | Optimization of the front end fast processing algorithms, based on analysis of each frame, alarm response 0.1S, no delay, no information loss |
| Visible camera        |                                                                                                                                               |
| Focal length          | 6.5 ~ 240mm, 37x optical zoom, 16x digital zoom                                                                                               |
| FOV                   | 1.86°~61.9°                                                                                                                                   |
| Defog                 | Electronic defogging / Optical defogging                                                                                                      |
| Focus                 | Manual focus/auto focus                                                                                                                       |
| Sensor type           | 1/1.8" Sony Starvis Progressive Scan CMOS Sensor                                                                                              |
| Total pixels          | 2688 × 1520, 4MP                                                                                                                              |
| Coding format         | H.265/H.264B/H.264M/ H.264H/MJPEG                                                                                                             |
| Network protocol      | IPv4, IPv6, HTTP, HTTPS, TCP, UDP, RTSP, RTCP, RTP, ARP, NTP, FTP, DHCP, PPPoE, DNS, DDNS, UPnP, IGMP, ICMP, SNMP, SMTP, QoS, 802.1x, Bonjour |
| API                   | ONVIF(Profile S, Profile G, Profile T), HTTP API, SDK                                                                                         |
| Thermal camera        |                                                                                                                                               |
| Focal length          | 50mm fixed zoom                                                                                                                               |
| FOV                   | 8.8°×7.0°                                                                                                                                     |
| Detector type         | Uncooled FPA                                                                                                                                  |
| Effective pixels      | 640×480                                                                                                                                       |
| Spectral range        | 8~14μm                                                                                                                                        |
| Temp sensitivity      | ≤35mK@F1.0,300K                                                                                                                               |
| Pan-tilt              |                                                                                                                                               |

# Dual Sensor Thermal Imaging System for Forest Fire Prevention

|                        |                                                                                         |
|------------------------|-----------------------------------------------------------------------------------------|
| Angle                  | Horizontal 360°continuous rotation, pitching -60°~+20°                                  |
| Speed                  | Pan: 0.01°~9°/s, Tilt: 0.01°~4°/s                                                       |
| Preset bit             | 255                                                                                     |
| Auto cruising          | 2 paths, 64 bits each path, support apple skin scanning(paths number can be customized) |
| Auto scanning          | 1 line                                                                                  |
| Angle return           | Support                                                                                 |
| Interface              |                                                                                         |
| Power supply interface | Width input waterproof power supply                                                     |
| Network interface      | 1×10Base-T/100Base-TX, RJ45                                                             |
| Other interface        | RS485                                                                                   |
| Environment indicators |                                                                                         |
| Working temp           | -25°C~+55°C                                                                             |
| Storage temp           | -40°C~+65°C                                                                             |
| Protection grade       | IP66                                                                                    |
| Physical features      |                                                                                         |
| Installation method    | Outdoor base installation                                                               |